📐 MATHEMATICS FOR MACHINE LEARNING

CampusX · Data Science Mentorship Program 2023 · Video 7 of 23

SESSION 43
Central Limit Theorem

Central Limit Theorem (CLT) · Sampling Distributions · Standard Error · Python Simulations

Introduction to Sampling | Sampling Distribution of the Mean | Central Limit Theorem (CLT) Formulation | Mathematical Proof and Intuition | The Role of Sample Size ($n \geq 30$) | Standard Error of the Mean | Real-World Importance in Machine Learning | Python Simulation of CLT

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)

📑 Table of Contents

01 · Introduction to Sampling

In statistics, analyzing an entire **population** is often impossible due to cost, time, or resource constraints. Instead, we select a representative subset called a **sample**.

A key objective of inferential statistics is to make claims about population parameters (e.g., population mean μ) using sample statistics (e.g., sample mean $\bar{X}$). Sampling distributions serve as the theoretical foundation for this process.

02 · Sampling Distribution of the Mean

If we take multiple random samples of size n from a population and calculate the mean of each sample, the probability distribution of these sample means is called the **Sampling Distribution of the Mean**.

As we take more samples, the histogram of sample means begins to show a distinct shape, centered around the true population mean.

03 · Central Limit Theorem (CLT) Formulation

The **Central Limit Theorem (CLT)** states that for any population with mean μ and standard deviation σ, the sampling distribution of the sample mean $\bar{X}$ will approach a Normal distribution with mean μ and standard deviation $\sigma / \sqrt{n}$ as the sample size n becomes large, **regardless of the shape of the original population distribution**.

CLT FORMAL MATHEMATICAL STATEMENT

As $n \to \infty$, $\bar{X} \sim N(\mu, \sigma^2/n)$

04 · Mathematical Proof and Intuition

The CLT is proved mathematically using Moment Generating Functions (MGFs). If we standardize the sample mean:

STANDARDIZED SAMPLE MEAN

$$Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$$

The MGF of Z approaches $e^{t^2/2}$ (the MGF of a Standard Normal distribution) as $n \to \infty$.

05 · The Role of Sample Size (n ≥ 30)

How large must n be for the CLT to apply?

- **Symmetric/Normal populations:** The sampling distribution is normal even for very small n (e.g., $n = 2, 5$).
- **Highly skewed/Non-normal populations:** A larger sample size is needed. The standard statistical rule of thumb is **$n \ge 30$**.

06 · Standard Error of the Mean

The standard deviation of the sampling distribution of the mean is called the **Standard Error (SE)**. It measures the variability of sample means around the true population mean.

STANDARD ERROR FORMULA

$$SE = \sigma / \sqrt{n}$$

As the sample size n increases, the standard error decreases. This means larger samples yield more precise estimates of the population mean.

07 · Real-World Importance in Machine Learning

CLT enables key data science tasks:

- **Hypothesis Testing:** Constructing test statistics for A/B testing.
- **Confidence Intervals:** Calculating bounds for model performance metrics (like mean absolute error).
- **Bootstrapping:** Sampling with replacement to estimate standard errors of model predictions.

08 · Python Simulation of CLT

```python
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generate skewed population data (Exponential distribution)
population = np.random.exponential(scale=2, size=100000)

sample_sizes = [5, 10, 30, 100]
fig, axes = plt.subplots(1, 4, figsize=(16, 4))

for idx, n in enumerate(sample_sizes):
    sample_means = [np.mean(np.random.choice(population, size=n)) for _ in range(1000)]
    sns.histplot(sample_means, kde=True, ax=axes[idx], color='#3b82f6')
    axes[idx].set_title(f'Sample Size n={n}')

plt.tight_layout()
plt.show()
```

09 · Quick Reference Card

🚀 Central Limit Theorem Cheat Sheet

- **Mean of $\bar{X}$:** Equal to the population mean μ.
- **Standard Error:** $\text{SE} = \sigma / \sqrt{n}$.
- **Shape:** Approaches Normal $N(\mu, \sigma^2/n)$ as $n \to \infty$.
- **Required sample size:** $n \ge 30$ is sufficient for most distributions.